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Dynamic multiphase DDoS attack identification and mitigation framework to secure SDN-based fog-empowered consumer IoT Networks

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ABSTRACT

The rapid growth of Consumer-Internet-of-Things (CIoT) devices has introduced critical vulnerabilities, making them susceptible to Distributed Denial-of-Service (DDoS) attacks. This paper proposes a multi-phase framework for detecting and mitigating DDoS attacks in Software-Defined Networking (SDN)-based fog-enabled IoT networks (SD-FCIoT). The framework combines entropy-based anomaly detection with machine learning for accurate and timely attack identification. Dynamic thresholding using Chebyshev's inequality ensures adaptability to network changes, while an attribute selection algorithm using symmetrical uncertainty and k-means clustering enhances classification accuracy. Experiments conducted in a simulated SD-FCIoT environment and with the BoT-IoT dataset demonstrate the framework's effectiveness. Random Forest (RF) achieved 98.71% accuracy for binary classification and excelled in multi-class attack detection, including Ping Flood and TCP-SYN Flood attacks. The attribute selection algorithm improved detection accuracy by 1.22% and reduced false positives. Furthermore, the fog-enabled architecture lowered response time by 35% compared to centralized systems. These results highlight the framework's scalability and efficiency in securing CIoT networks against diverse DDoS attacks.

1. Introduction

Evolution in wireless technologies and communications embellishes the ubiquitous network of heterogeneous "things", nowadays entitled the Internet-of-Things (IoT) network, where devices function collaboratively to accomplish a fixed set of tasks [1,2]. IoT-based solutions are interwoven into nearly every domain ranging from smart home to critical infrastructure systems [3]. The growth of IoT applications presents significant market opportunities for device vendors. The unsupervised working environment, resource-constrained, and insufficient processing capabilities hinder the security of IoT devices [4]. Additionally, manufacturers prioritize infrastructure growth over device safety, resulting in significant security vulnerabilities [5–8] that manifest numerous attacks like Distributed Denial of Service (DDoS) attack.

DDoS erupts from a network of malign devices, known as the botnet, that a malicious actor controls with a mission to thwart the victim's online services to legitimate actors [9]. To accomplish this goal, the malicious actor engages in a technique known as

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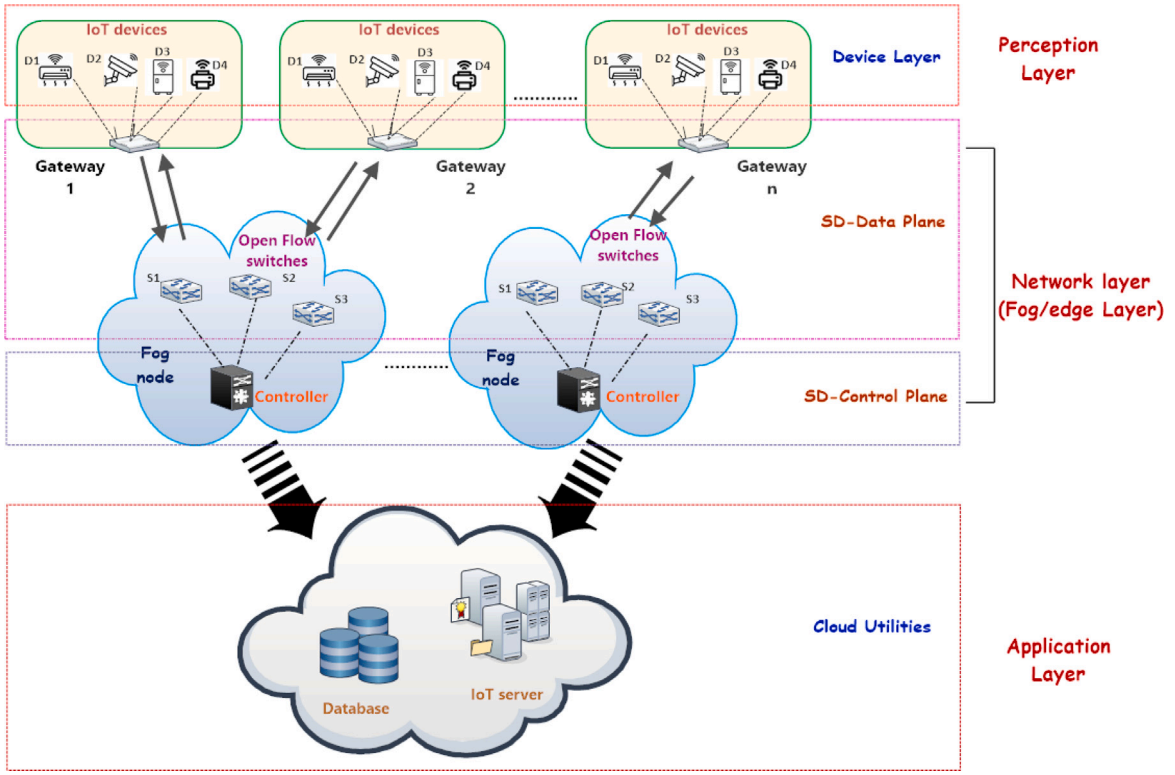


Fig. 1. Layout of SD-FCIoT System.

packet flooding, wherein an excessive number of spurious packets are directed toward the victim. This influx of packets overwhelms the victim's processing and storage capacities, ultimately culminating in a system failure [10]. According to the projections [11], the global count of interconnected devices is anticipated to surpass 29 billion by the year 2030. Moreover, in 2020, the consumer sector is estimated to embrace almost 60% of the total IoT devices which leads to the emergence of Consumer-Internet-of-Things (CIoT). If these devices are supplied with latent vulnerabilities that were bypassed in quality assurance, then these devices possess the capacity to destroy privacy and data integrity, or even imperil the internet itself [12]. It is arduous to control, manage, and secure such a vast network of heterogeneous devices. Hence, the IoT network is amalgamated with the recent networking architecture approach i.e., Software Defined Network (SDN). It provides numerous benefits to the IoT networks, including centralized control of the network to disperse security policies, flexible and effective management of devices, efficient resource sharing, and intelligent routing, resulting in increased agility and scalability of the network [13,14]. Furthermore, the deployment of conventional security mechanisms in centralized IoT infrastructure pose several issues comprising bandwidth exhaustion, scalability and mobility issues, high latency, resource inadequacy and reduced Quality of Service (QoS). Therefore, fog computing emerges to solve these problems through processing huge volume of data near to end devices [15] In this work, authors propose to utilize the capabilities of SDN-controller in fog-enabled consumer IoT network. Fig. 1 depicts the architectural integration of SDN, fog computing and IoT network (SD-FCIoT). In this framework, the SDN-controller acts as the control plane in the fog node and provides the centralized control of the IoT network whereas, data plane is accountable for forwarding the packets.

Conventional DDoS defensive techniques solely rely either on statistical analysis or machine learning. Firstly, statistical analysis-based techniques depend on a predefined variable i.e., threshold to identify attack traffic. When the observed statistics surpasses the threshold then it indicates some anomalous activities. However, threshold selection is affected by various factors and is subjective in nature. Moreover. if the threshold range is set too large then some attack traffic may pass unnoticed, otherwise, if it is set too low then normal traffic may be seized that affects the services to the users. Thus, these techniques are inefficient and possess low attack detection accuracy. Secondly, machine learning-based approaches [16] incurs high computational overhead and detection latency that leads to untimely detection of malicious traffic. Additionally, these techniques are resource intensive and are sensitive to the noisy attributes, therefore, it is not efficient to execute ML model on controller consistently for attack detection. Thirdly, it is also imperative to devise a mitigation method to filter the malicious traffic carefully based on detection outcomes without hampering the legitimate traffic.

Our work: Motivated by these shortcomings, authors have proposed an attack detection and mitigation framework through leveraging the benefits of SDN in the fog nodes of consumer IoT network. It is a multi-phase framework which amalgamates the benefits of statistical analysis and machine learning for early identification and multi-class classification of network traffic.

First phase applies information entropy method to identify the abnormal activities and filter out the suspicious connections in the network. Here, threshold value is dynamically calculated based on the traffic statistics using Chebyshev's inequality [17], to reduce the false positive alarms. Second phase initially implements the proposed feature selection algorithm which is based on symmetrical uncertainty and K-means clustering methods. Then, it employs machine learning models comprising Support Vector Machine (SVM), Random Forest (RF), J48, K-Nearest Neighbor (KNN), and Extreme Gradient Boosting (XGBT), for multi-class classification of the attack traffic. Third phase mitigates the attack through transforming the flow policies based on the detection results. This integrated framework ensures efficient and timely detection of the attack traffic. The key contributions are outlined as follows:

- Proposed a triple-phase DDoS attack detection and mitigation framework based on information entropy and machine learning for protecting consumer devices in SD-FCIoT network.
- Proposed an attribute selection algorithm based on the Symmetrical Uncertainty and K-means clustering for selecting the prominent attributes that aids in enhancing the detection efficiency of classifiers.
- ·Assess the performance of various machine learning classifiers including Radial Basis Function SVM (RBF-SVM), RF, J48, KNN and XGBoost for binary and multiclass classification.
- Proposed a mitigation method by incorporating or transforming flow policies at every switch in the network based on the suspicious links and devices, for hamper attack traffic.
- Setup a simulation environment in Mininet, to test the real-time attack detection efficacy of the proposed framework. The attack traffic is generated for various classes including Ping flood, UDP flood, SYN flood, HTTP flood, and NTP-amplification. Moreover, authors evaluated the performance of proposed framework on Bot-IoT dataset to reflect the robustness and effectiveness of the work.
- Measure the performance of each classifier in terms of accuracy, false positive rate, misclassification rate, precision, recall, and F1-score for both the cases i.e., with and without attribute selection using K-fold cross validation method for binary as well as multiclass classification.
- Response time of best performing classifier is estimated for both centralized and fog-enabled IoT network.

This article is schemed as follows: Section 2 shed some light on the key contributions of the researchers in the related field. Section 3 represents the attack model in the SD-FCIoT network. Section 4 demystifies each phase of the proposed framework, followed by the discussion on simulation topology and experimentation outcomes in Section 5. Finally, Section 6 concludes the research with future prospectives.

2. State-of-art defensive techniques and motivation

This section demystifies the key contributions of the different academicians and researchers in the related field. It also illuminates the motivation of carrying out this research work.

2.1. Existing defensive techniques

Many defensive solutions are proposed by different researchers to defend against DDoS attacks. There are multiple classes of DDoS attacks and several studies have designed intrusion detection system (IDS) [18,19].

2.1.1. Information entropy-based solutions

Basicevic et al. [20] examine the integration of principal component analysis (PCA) with entropy-based approaches for identifying DDoS attacks. They examine Shannon entropy, Tsallis entropy, and Bhatia-Singh entropy. It is unveiled in this study that the use of PCA with entropies, specifically Tsallis and Bhatia-Singh, outperforms entropy-based approaches in terms of metrics including recall, F1-score, and FPR. Gaurav et al. [21] propose a method for identifying DDoS attack at the fog layer in the IoT network. The Fog node computes the entropy of incoming traffic by considering the SIP address as a random variable. An attack is identified when the level of entropy exceeds a predetermined threshold value. Bagla and Kumar [22] introduces the TA-WHI model for classifying online health information from social media into five categories: sufficient, fabricated, meaningful, advertisement, and misleading. Ali et al. [23] design a method by integrating entropy and sequential probability ratio test (SPRT) to identify DDoS attack. The entropy ascertains randomness in destination IP addresses, which serves as the input for the SPRT. The outcome is compared to both a lower and a higher threshold to identify DDoS attacks. Nevertheless, these threshold values remain fixed and do not monitor fluctuations in network traffic. Lai and Zeng [24] introduces a POI recommendation algorithm using an attention mechanism combined with LSTM to capture sequence characteristics and short-term preferences. Dung et al. [25] explores the convergence of IoMT and blockchain technology, highlighting its potential to revolutionize personalized healthcare by enabling secure and efficient systems. Li et al. [26] proposed a method to identify volumetric DDoS traffic based on joint-entropy approach. It has a traffic processor, entropy calculator, and detecting module. The traffic processor analyzes network traffic, preprocesses data, and sends it to the entropy calculator which estimates joint entropy for detection module. The detection module compares entropy values to risk models and determines alert issuance. Authors tested the efficiency on three standard datasets. Due to static threshold, this scheme cannot adjust with network changes. Gupta et al. [27] presents the W-RLG Model, a deep learning framework combining Whale Optimization, RNN, LSTM, and GRU for detecting cyber threats in healthcare IoT systems. Mishra et al. [28] proposed a DDoS identification and mitigation method for SDN environment. They use POX controller for monitoring network activities. This method utilizes entropy variations for normal and attack traffic with low overhead. It achieves a detection rate of 98.2%. Nevertheless, it is speculated that these approaches

utilize a predefined threshold value which has a significant impact on attack detection during network fluctuations. Moreover, to set threshold in different scenarios requires a lot of experience and expertise, that may directly affect the detection accuracy. Balraj et al. [29] proposes the Blockchain Interplanetary File Access Control System (BIFACS) to address security, transparency, trust, and data management issues in IoT-enabled Wireless Sensor Networks (IWSN). Alhalabi et al. [30] provides a survey of IoT-based DDoS detection methods, focusing on neural networks and proposing a decision tree-based model for detecting DDoS attacks.

2.1.2. Machine learning-based solutions

Machine learning-based attack detection is a cornerstone in IoT network [31]. Kumar et al. [32] designed an IDS for blockchain-based IoT network to identify DDoS attack. They use federated learning to train the global detection model and disseminate it to each fog node in the network. Thus, each fog node has the capability to identify and detect DDoS attack. They conducted experiments using the BoT-IoT dataset to evaluate the performance of random forest and XGBoost classifiers. The findings indicate that the distributed training procedure may effectively decrease the duration of the training phase. Mehmood et al. [33] introduced an IDS based on a multi-agent system framework. This system utilizes the Naïve Bayes classification technique to identify abnormalities in the network. Roopak et al. [34] examined the effectiveness of deep learning models, such as multilayer perceptron, 1d-Convolution Neural Network, Long Short-Term Memory, and CNN + LSTM, in detecting attacks. They used the CICIDS2017 dataset for their analysis. Dao et al. [35] introduced MECShield, a platform that utilizes mobile edge computing to identify attacks in local IoT networks using smart filters (SOM). Soe et al. [36] developed an IDS and evaluated its performance on a Raspberry Pi device. A novel feature selection approach, CST-GR, was developed to identify significant characteristics from the Bot-IoT dataset. The detection effectiveness of tree-based classifiers including J48, VFDT, LMT, and RF has been evaluated. Roopak et al. [37] Developed a feature selection approach using multi-objective optimization to identify attacks in IoT networks, using the ELM classifier. The feature selection approach combines the Jumping gene modified NSGA-II algorithm with 6 major objectives.

Numerous techniques are designed based on the integration of several methods. Jain et al. [38] surveys blockchain consensus algorithms, comparing their advantages and limitations, and covers both traditional and emerging protocols. Numerous techniques are designed based on the integration of several methods. Sahoo and Puthal [39] designed a framework for identifying DDoS attack in IoT network consisting of multimedia devices by using SDN capabilities. They use entropy and machine learning models including NB, SVM, decision tree, and RF to classify the attack traffic. To optimize classifier's performance, they use Adaboost optimization technique. Prasanth et al. [40] provides a comprehensive overview of the convergence of Internet of Medical Things (IoMT) and blockchain technology, focusing on their applications in personalized healthcare, security, and privacy. [41] introduces an Energy-Aware Hierarchical Rule-based Software Fault Detection (HRSFD) model for IoT-enabled Wireless Sensor Networks (IWSNs). Ali et al. [42] designed a framework for intrusion detection in IoT-based maritime transportation system. It uses 10 tuple-based entropy along with rate of exponent separation. It analyzes the performance of MLP, logistics regression, and alternating decision tree on MTS-IoT dataset. This comprehensive methodology illuminates a new way for timely detection and mitigation of attack in IoT networks. Nevertheless, there are issues to be focused upon. Firstly, the use static threshold affects the detection accuracy of the system under changing network behavior. Secondly, most of the approaches are focused on attack detection and less priority is paid to the resolution. Moreover, we can reduce detection latency through incorporating defense at the edge. Therefore, authors have designed a framework to secure consumer devices by harnessing the capabilities of SDN in the edge of the IoT network.

2.2. Motivation

Recent research has unveiled multiple flaws in IoT networks. Certainly, CIoT devices are captivating "bots" in the hand of an attacker because of several reasons. Firstly, these devices are currently in high demand due to their compact size and lower cost, resulting into huge volume of network traffic. Secondly, resource constrained nature hinders the incorporation of defensive solutions. Thirdly, these devices utilize wireless communications, and hence, can be easily discovered on the public network. Moreover, due to latent web application flaws such as cross site scripting (XSS) [43], these devices are prone to malware infections and become part of botnet, resulting into DDoS attack at massive scale.

In the realm of IoT, malicious traffic often combined with voluminous benign traffic, making it a cumbersome task to differentiate at the cloud level. Therefore, it is crucial to ensure that the attack is thoroughly examined at the edge level. To overcome this problem, the authors propose the deployment of SDN in fog nodes as a crucial solution. Many defensive solutions primarily focus on utilizing either machine learning or statistical techniques to detect attacks. Nevertheless, Alsmadi and Xu [44] demonstrated that while ML techniques are promising, however, it is not advisable to repeatedly execute the classifier for detecting attack traffic in large networks. Thus, authors have designed a framework that use statistical analysis and machine learning to distinguish attack traffic and mitigate its effects. Offering a SD-FCIoT architecture is an innovative idea that can effectively handle the influx of DDoS traffic caused by a vast amount of data originating from the consumer IoT devices.

3. Network architecture: Attack model

Consumer IoT sector is the most widely spread IoT network. Smart consumer devices are more fascinating to the attacker due to their extensive usage in uninterrupted and poor maintenance environment. Fig. 2 provides a clear illustration of an attack scenario in SD-FCIoT network. It is considered that an adversary originates from outside, whereas the SDN controllers in the fog node are trustworthy and intact. It may be summarized as follows:

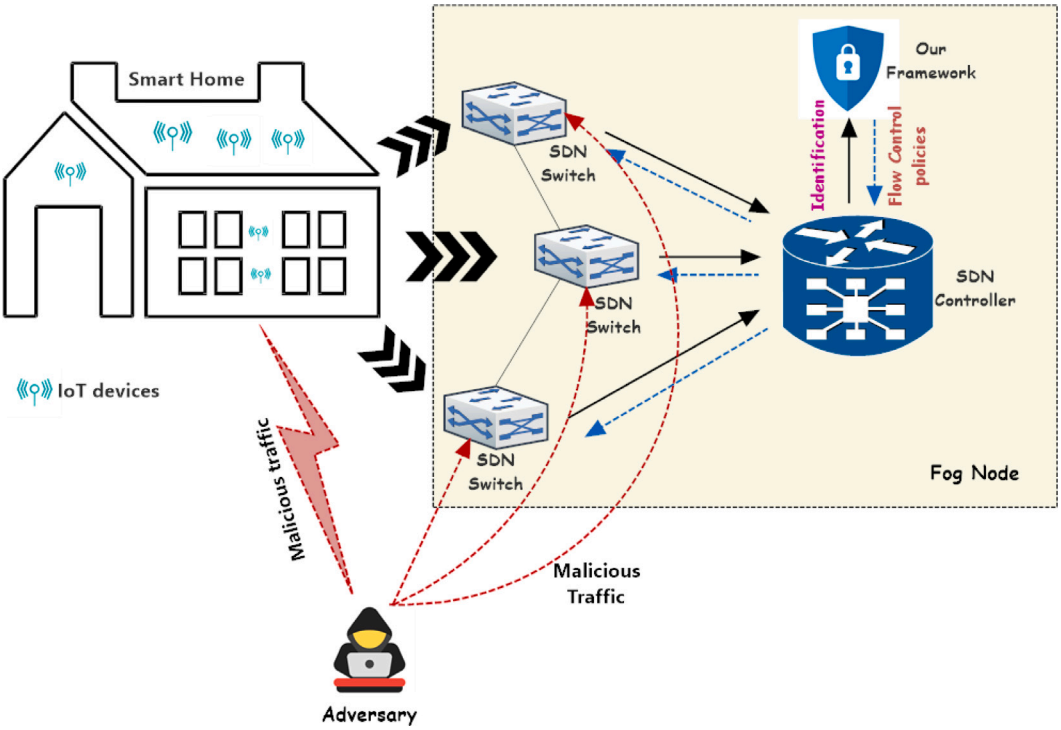


Fig. 2. An instance of attack scenario in SD-FIoT.

- · ***Types of attack:*** DDoS attack encompasses a wide range of attacks. One of the most prevalent attacks is the flooding attack in which an attacker deliberately overwhelms the victim with an excessive amount of network traffic [45]. It results into the exhaustion of the victim's resources or disruption of the network infrastructure. In this research, authors focus on UDP flood, ping flood, TCP-SYN flood, and HTTP flood amongst flooding attacks. Additionally, authors consider the NTP (Network Time Protocol) amplification attack that takes illegitimate advantage of NTP servers to influx victim with unsolicited messages that eventually results into victim's crash.
- ***Attacker Proficiency:*** It is facile to launch a DDoS attack using hacked consumer IoT devices since many are susceptible to infiltration. An attacker proficiency includes the following:
  - a. Attacker may exploit the latent security vulnerabilities in the consumer device's firmware to create a botnet army.
  - b. Attacker may trigger multiple known and/or unknown DDoS attacks using the created botnet.
  - c. Attacker may use spoofing to hide its identity or to be difficult to recognize from the pool.

4. Proposed framework

The proposed framework manages and secures CIoT nodes at the edge level using OpenFlow protocol [46]. This framework is merged with the controller to accomplish the designated task. This section comprehensively describes the functional phases comprising attack identification, classification, and mitigation. In the following section, fundamental methodologies are discussed in brief and precise manner.

4.1. Methodologies: Fundamental concepts

This section summarized the methodologies used in this research work.

A. Information entropy: Information theory uses entropy to assess uncertainty [47] Lesser entropy correlates with lesser information uncertainty, and vice versa. In flooding attacks, packet contents are substantially copied, reducing entropy. Consider, x, a discrete random variable which accepts value in the alphabet and is distributed based on a probability : $P : X \rightarrow [0, 1]$ such that p(a) = P[= a]. Then, Shannon entropy of x, denoted as H(x), can be represented by the following equation:

$$H(X) = -\sum_{a \in X} p(a) log_2 p(a) \quad (1)$$

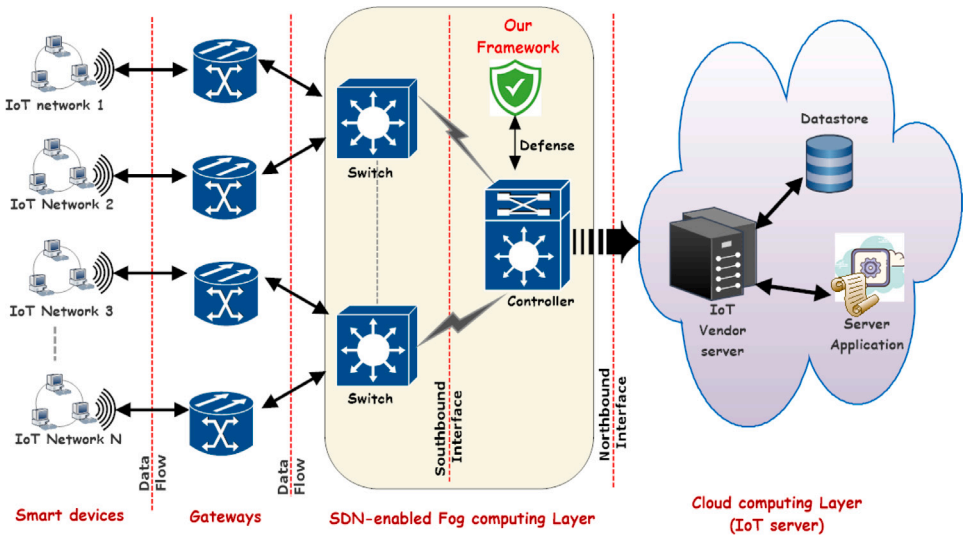


Fig. 3. Abstract Design model of the proposed framework.

Then, the normalized entropy can be calculated as follows:

$$E_{nor} = -\frac{1}{log_2|X|}\sum_{a\in X} p(a)log_2 p(a) \tag{2}$$

Here, the E_{nor} lies in the interval [0,1].

B. Symmetrical uncertainty (SU): It is a metric to measure the correlation between two random variables [48]. Let X and Y be the discrete random variables then information gain is defined as follows:

$$IG(X|Y) = H(X) - H(X|Y) = H(Y) - H(Y|X) \tag{3}$$

where, H(.) denotes the entropy and H(X|Y), or H(Y|X) denotes the conditional entropy which is defined as follows:

$$H(X|Y) = \sum_{a\in X}\sum_{b\in Y} p(a,b)log\frac{p(a,b)}{p(a)p(b)} \tag{4}$$

where, a, b denotes the possible observed values of X and Y. p(a), p(b) and p(a, b) indicates probability mass distribution of a, b, and their joint probability distribution, respectively.

IG tends to prioritize variables that have multiple values. SU normalizes the value of IG and sets its range to [0,1]. It is represented by the following equation:

$$SU(X,Y) = \frac{2 * IG(X|Y)}{H(X) + H(Y)} \tag{5}$$

Here, SU is used to rank the attributes based on its relevancy to the class variable (i.e., normal, or malicious) that helps in reducing the effect of redundant or noisy attributes on classifier accuracy.

4.2. Abstract design schema

In conventional DDoS methods, the target environments are usually responsible for detecting and mitigating attacks, which is a reactive approach. Nevertheless, the SD-FCIoT architecture provides the possibility of transforming this paradigm into a more distributed, and "closer-to-end-nodes" shielding architecture in an IoT context. Due to low detection rate, high latency, and scalability challenges, it is advised to defend closer to the infected nodes i.e., at edge, particularly for DDoS attack. It employs the fog computing paradigm which is an extended version of cloud computing at the edge of the network with all cloud services including computation, storage, and communications. Thus, the main function of fog computing is to harness the capabilities of SDN controller at the edge to ensure security. Fig. 3 highlights the abstract design of the proposed framework to identify and mitigate the DDoS attack. Infected smart home devices are linked to the SDN-controlled fog network. There may be multiple fog nodes in the network for providing services to the user at reduced cost, latency, and location awareness property.

It consists of 3 key functional phases, namely, anomaly identification, attack classification, and policies reformation. In the first phase, statistical analysis is used for early alarm to the anomalous activities in the network. To achieve this, authors implement Shannon entropy on traffic attribute as it measures the randomness of a variable. To adapt to the changes in the network traffic, we dynamically set the threshold value by using Chebyshev's inequality. This helps in overcoming the challenges faced due to

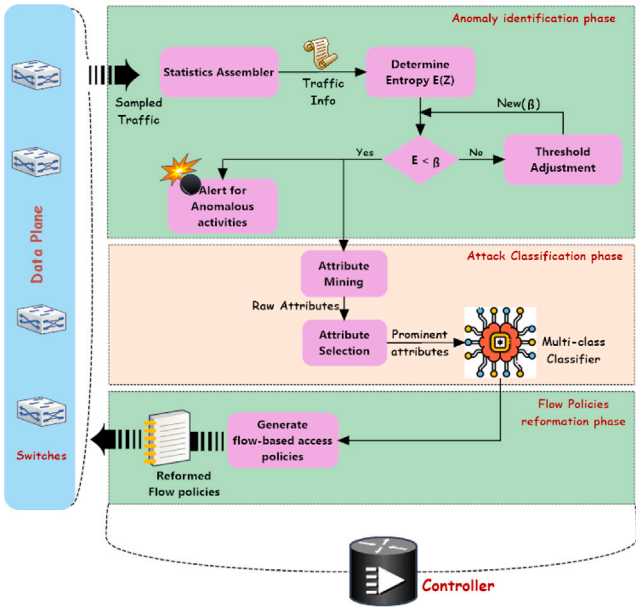


Fig. 4. Detailed Design model of proposed framework.

static thresholds. In the second phase, firstly, authors filter the noisy attributes through providing ranking to the attributes based on symmetrical uncertainty method. It helps in enhancing the classifier's accuracy. Secondly, multi-class classification is performed to unveil distinct categories of the DDoS attack in the network. In the last phase, based on the detection outcomes, authors reform the flow policies at each switch so that they may act as a filter to malicious traffic. Authors utilize OpenFlow 1.5.1 protocol-based controller to accomplish the desired task.

4.3. 1.1. detailed design schema

Fig. 4 shows the detailed modular design of the proposed attack defensive framework for SD-FCIoT environment. Here, it is assumed that the framework takes sufficient time for building the system profile under normal scenario. In this section, initially, authors describe the workflow of the proposed framework. Then, each module is defined, in detail, in the following sub-sections.

4.3.1. Functional-flow

The functional flow of the proposed framework consists of the 3 phases: anomaly identification, attack classification, and policies reformation. Fig. 5 represents the comprehensive workflow of the proposed framework.

A Identification: It collects the network statistics such as source IP address, destination IP address, port, protocol, etc. through *flow_mod* message and estimate normalized entropy (E_{nor}). If this entropy is lower than the threshold then it indicates toward anomalous activities in the network.

B Classification: It is launched when there an alarm is generated by the previous phase. It performs attribute selection to highlight prominent attributes that are used for attack classification. Moreover, it employs multiple machine learning classifiers to achieve multi-class classification to represent distinct classes of the DDoS attack.

C Flow-policies reformation: It is designed to apply mitigation policies after attack detection at the switches. Initially, the infected devices are identified by using link frequency and then policies are formed that may turn a switch into a filter for the attack traffic. These policies are forwarded to the switch via *flow_mod* message.

4.3.2. Functional phases

Here, we exhaustively discuss the 3 phases of the proposed framework.

A. Anomaly Identification: This phase is based on the statistical analysis for early identification of the anomalous activities in the network. It involves following modules:

i. *Network statistics assembler:* This module generates bidirectional links (L) with their link frequency from network traffic at a regular interval of time frame. Link is determined by the pair of IP addresses <*SIP, DIP*> which are extracted from packet header. We focus on 3 aspects of the link: the outgoing packet frequency (LF_{OUT}), the incoming packet frequency (LF_{IN}), and the timestamp of the initial packet in link *(TS)*. Furthermore, it extracts DIP into a hash table and estimate its occurrence frequency (χ) as $\mathbf{h}(D_p, \chi)$ Algorithm 1 defines each step of this module.

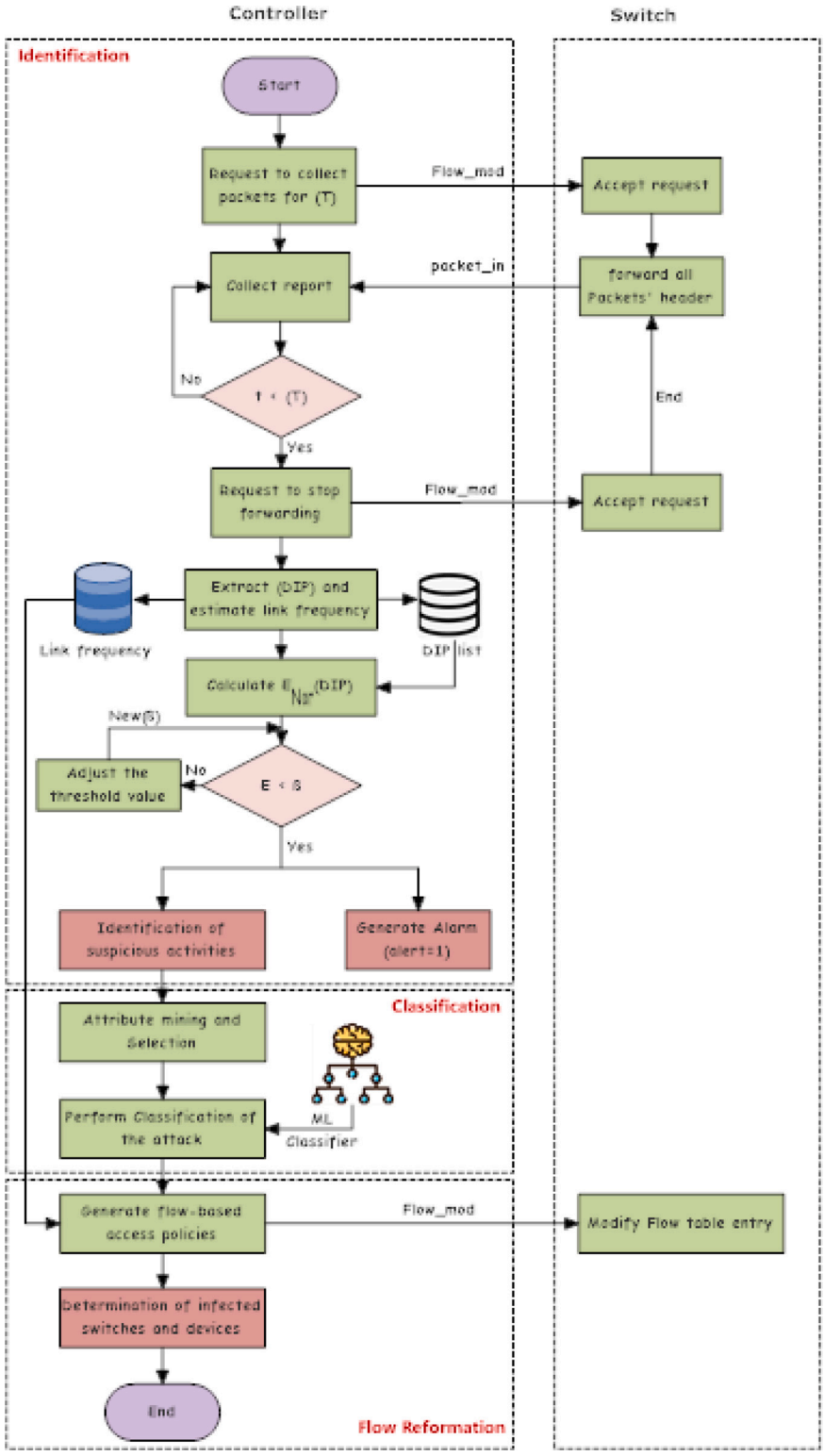


Fig. 5. Operational work-flow of the proposed framework.

Algorithm 1 Network Traffic Statistics Assembler

Require: Network traffic, Time frame (ΔT)
Ensure: $\hbar(D, \chi)$, Links (L), Link Frequency (LF_{IN}, LF_{OUT})
$S \leftarrow$ Set of switches
$L \leftarrow []$ ▷ List of links in the network
$T_{ini} \leftarrow$ Initial time
$T_{pre} \leftarrow$ Present time
$\hbar(D, \chi) \leftarrow$ Destination IP hash table with χ occurrences
$LF_{IN} \leftarrow 0$ ▷ Incoming link frequency
$LF_{OUT} \leftarrow 0$ ▷ Outgoing link frequency
while $T_{pre} - T_{ini} < \Delta T$ **do**
  for each $s_i \in S$ **do**
    Collect traffic statistics of s_i ▷ Collect source and destination IPs with timestamp
    $P_i \leftarrow$ Extract packet attributes (SIP, DIP, TS)
    if $P_i(SIP, DIP) \in L$ **then**
      $LF_{OUT} \leftarrow LF_{OUT} + 1$
    else if $P_i(\neg DIP, SIP) \in L$ **then**
      $LF_{IN} \leftarrow LF_{IN} + 1$
    else
      $L \leftarrow L \cup P_i(SIP, DIP)$
      $LF_{OUT} \leftarrow LF_{OUT} + 1$
    end if
    for each $P_i(DIP)$ **do**
      if $P_i(DIP) \in \hbar$ **then**
        $\hbar(D, \chi) \leftarrow \chi + 1$ ▷ Where $D = DIP$
      else
        $\hbar(D, \chi) \leftarrow 1$
      end if
    end for
  end for
end while
return L, LF_{IN}, LF_{OUT}, $\hbar(D, \chi)$

i. *Entropy determination and alert generation:* It takes link information, and DIP table generated in previous module along with threshold β and ω, as inputs and returns alert to the anomalous activities in the network. The randomness in the *DIP* address during the attack is deemed to be moderate because traffic is generated for a specific victim. Normalized entropy (E_{nor}) is calculated by considering *DIP* as the random variable. For time frame, the probability distribution p(.) of *DIP* is estimated as:

Algorithm 2 shows the functions of this module. A lower normalized entropy value of less than β generates the alert to malicious activities which acts as a signal to trigger attack classification module.

Algorithm 2 Entropy Determination and Alert Generation

Require: Link (L), $\hbar(D, \chi)$, Threshold (β)
Ensure: Alert for anomalous activity
$Alert \leftarrow 0$ ▷ Boolean variable initialized to 0 (false)
$E_{nor} \leftarrow 0$ ▷ Normalized entropy
$M_{link} \leftarrow []$ ▷ List of illegitimate links
for each $h_i \in \hbar$ **do**
  $E_{nor} \leftarrow$ Estimate normalized entropy (D)
  if $E_{nor} < \beta$ **then**
    $Alert \leftarrow$ Generate alert signal (1) for anomalous activity
    Trigger Multiclass Attack Classification module()
    Trigger Threshold Adjustment module()
  else
    No alert
    Network is safe
  end if
end for

iii. *Threshold Adjustment:* This module is designed to adjust the threshold value as per the changes in the network traffic. It adjusts two thresholds, β for anomaly detection, and ω for suspicious devices identification, based on the Chebyshev's inequality. It is a fundamental concept in probability theory which ensures that, over a broad range of probability distributions, the proportion of values that deviate from the mean by a particular distance is limited. It can be stated that at least a fraction of (1 - $1/k^2$) of the values in the distribution are fewer than 'k' standard deviations away from the mean. Technically, consider 'X' as a random variable with finite non-zero variance (σ^2) and mean (μ) then for any random variable k> 0,

$$p(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2} \tag{6}$$

In this work, k = w. In simple terms, re-write above equation as:

$$[\mu \pm k\sigma] \geq \left[1 - \frac{1}{k^2}\right] \tag{7}$$

Authors select lower bound of the interval as threshold 'β' because during the attack, randomness in DIP decreases which leads to lower entropy. On the other hand, there is a substantial increase in the number of attack packets during flooding attack, therefore, authors select 'ω' as upper bound of the interval. Algorithm 3 shows the steps taken for this module. At the initial stage, we adjust threshold value based on the normal scenario. Here, authors sample traffic for a time frame of ΔT and focus on '$\Re$' values. Then, compute normalized entropy vector (ENV) and extract out the last $\Re$ values of the average number of packets in normal link (Lavg). Finally, threshold values are computed as below:

$$\beta = \mu = w\sigma \quad and \quad \omega = \mu + w\sigma \tag{8}$$

B. Attack Classification: This phase efficiently performs the multi-class classification of the malicious traffic to distinguish various attack categories. To achieve this objective, it consists of following modules:

Algorithm 3 Threshold Adjustment

Require: Normalized entropy vector (ENV), Time frame (ΔT)
Ensure: Revised threshold (β)
1: $\mathbb{R} \leftarrow$ Random number
2: $ENV \leftarrow []$
3: $\omega \leftarrow$ Non-zero natural number
4: **while** $\mathbb{R}$ **do**
5: $e_i \leftarrow$ Calculate entropy of traffic for ΔT
6: $ENV \leftarrow ENV \cup e_i$
7: **end while**
8: $L_{avg} \leftarrow$ Last $\mathbb{R}$ values of the average number of packets of normal link (L)
9: $\mu \leftarrow$ Calculate mean of ENV
10: $\sigma \leftarrow$ Calculate standard deviation of ENV
11: $\hat{u} \leftarrow$ Calculate mean of L_{avg}
12: $x \leftarrow$ Calculate standard deviation of L_{avg}
▷ Estimate threshold value as the lower and upper bound of Chebyshev's inequality
13: $\beta \leftarrow \mu - \omega \cdot \sigma$
14: $\omega \leftarrow \hat{u} + \omega \cdot x$
15: **return** β, ω

i. Attribute mining and selection: This module is responsible for eliminating noisy attributes that helps in enhancing classifier's accuracy. There are 2 key objectives for designing an attribute selection algorithm:

- *Objective 1:* To identify the most prominent attributes. For this, authors will calculate the relevance score of each attribute with the class variable which defines attribute–class relevancy. The relevancy score is estimated by using $SU(a_i, T_C)$, where 'a_i' indicates an attribute and 'T_C' denotes class variable.
- *Objective 2:* To eliminate noisy attributes. For this, authors will estimate the redundancy matrix (M_{SU}) based on the redundancy score between the selected attribute and the remaining set of other attributes which defines attribute–attribute redundancy. The redundancy matrix is designed by using $SU(a_i, a_j)$, where 'a_i' and 'a_j' are two different attributes of the dataset.

Algorithm 4 represents the operations of this module. It accepts raw attribute (A) as input and return optimal attribute set (A_{min}) as output. Here, after calculating M_{SU}, authors apply k-means clustering method [49] to perform attribute grouping (G_A) based on the redundancy score. Each group is representing a subset of attributes that share approximately similar relevancy score with the class variable. To choose optimal set of attributes (A_{min}), authors choose only a representative attribute from each group as all other

attributes are redundant. This selection is based on the following equation.

$$\underset{a_1}{argmax}\left(SU(a_i,T_c)-\frac{1}{|g|}\sum_{a_j\in g}SU(a_j,a_i)\right) \tag{9}$$

where, 'ai' is the representative attribute that has the highest SU score. This process iterates until we select one attribute from each group.

Algorithm 4 Attribute Mining and Selection

Require: Raw attribute set $A=\{a_1,a_2,\ldots,a_x\}$
Ensure: Optimal attribute set (A_{min})
$A_{min} \leftarrow []$ ▷ Optimal attribute set
$TC \leftarrow []$ ▷ Target labels
$MSU \leftarrow []$ ▷ $x \times x$ matrix
$GA \leftarrow 0$ ▷ Number of attribute groups
$SU \leftarrow []$ ▷ Symmetrical uncertainty vector
for each $a_i \in A$ **do**
    $SU_i \leftarrow$ Estimate $SU(a_i,TC)$, where $i \in \{1,2,\ldots,x\}$
end for
$A_{min} \leftarrow$ Arrange attributes in descending order of SU ▷ Compute SU matrix
for $\forall i,j \in \{1,2,\ldots,x\}$ **do**
    $w_{ij} \leftarrow SU(a_i,a_j)$
    $MSU \leftarrow w_{ij}$
end for
▷ Implementing k-means clustering on attribute set
$GA \leftarrow$ Apply k-means clustering on MSU
for each $g \in GA$ **do**
    Select attribute a_i:
    $a_i \leftarrow \arg\max_{a_i}\left(SU(a_i,TC)-\frac{1}{|g|}\sum_{a_j\in g}SU(a_j,a_i)\right)$
    $A_{min} \leftarrow a_i \cup A_{min}$
end for
return A_{min}

ii. Multi-class classification: This module receives the optimal set of attributes as input and provides the classification results that shows traffic proportion of each attack category in the network. It utilizes machine learning classifiers to distinguish between regular network traffic and malicious attack traffic. The process include training or constructing the model, which is then evaluated against real-time data to identify patterns of attack traffic. Authors are aiming at the designing of a lightweight classification module; thus, employ least complicated classifiers including SVM, RF, J48, KNN, and XGBT. The classifier's performance is evaluated based on predefined benchmarks, as explained in Section 5.3. Finally, the detection outcomes are forwarded to the next module for network policies reformation at switch-level that aids in filtering out the malicious traffic. Algorithm 5 depicts the steps for this operation.

Algorithm 5 Multiclass Attack Classification

Require: Alert, Optimal attribute set (A_{min})
Ensure: Attack classification results ($\hat{R}$)
$C \leftarrow$ List of ML classifiers
$\hat{R} \leftarrow \emptyset$
$D_{A_{min}} \leftarrow$ Dataset with optimal attributes
for each $c_i \in C$ **do**
    Initialize c_i
    Training: $c_i \leftarrow D_{A_{min}}$ ▷ Train the model with standard dataset
end for
if Alert **then**
    Classification: $c_i \leftarrow Dataset(A_{min})$
    Classify the attack type
    $\hat{R} \leftarrow$ Results of c_i
    Mitigation: Trigger Rules and Policies generator module()
end if
return $\hat{R}$

C. Flow-based Access Policies Reformation: This module is designed to apply mitigation policies to the switches that helps in defending against the attack. This mitigation method is unaffected by the SDN's topology and is capable of handling intricate and unfamiliar attack scenarios. After attack detection, it is mandatory to reform the flow policies so that the malicious traffic gets

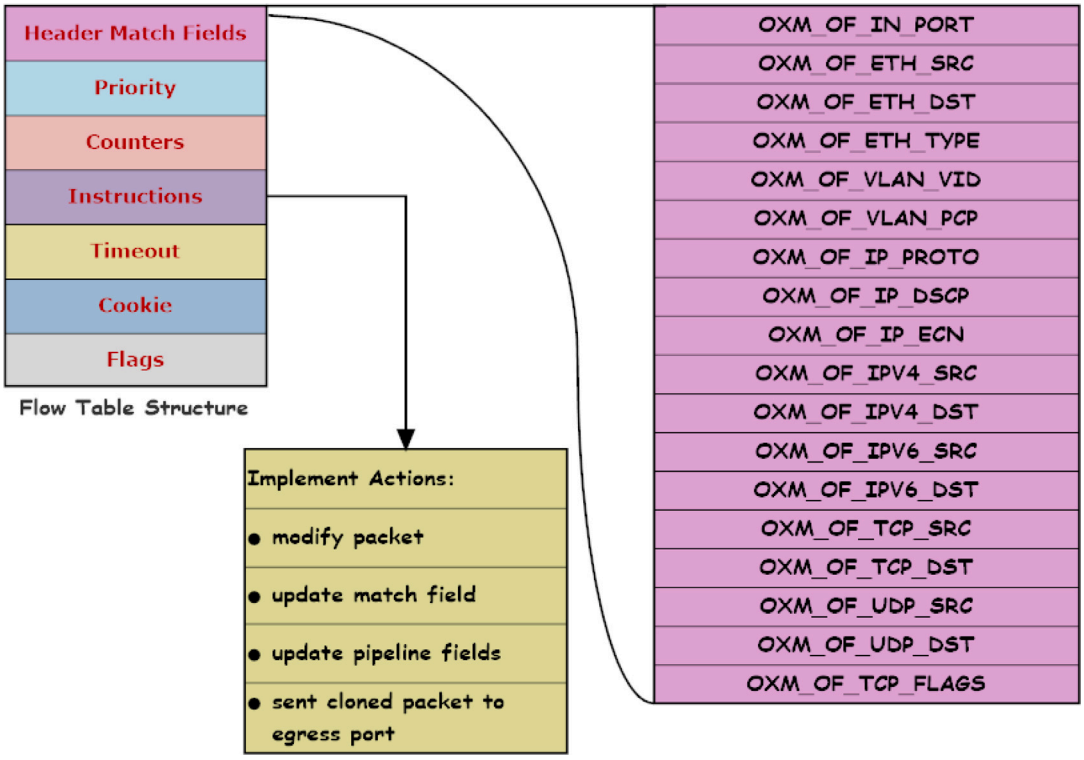


Fig. 6. Openflow 1.5.1 flow-table structure.

Table 1
An instance of flow table entry to mitigate NTP amplification attack.

IP_SRC	IP_DST	UDP_SRC	UDP_DST	IP_PROTO	Action
10.0.0.1	10.0.0.3	11 487	123	UDP	DROP
10.0.1.2	10.0.0.3	1234	8888	UDP	PASS

scrutinized instantly. Flow policies are modified as per the specifications provided in the OpenFlow 1.5.1 [50]. Here, authors choose OpenFlow-hybrid switches and utilize a total of 18 "match fields" in the flow tables to match against network packets, as shown in Fig. 6.

For instance, OXM_OF_TCP_FLAGS field can be matched to identify SYN, ACK, and FIN bit to detect the beginning and ending of a TCP connection. Here, instructions can be modified to guide what actions are to be performed when a packet matches with any entry in flow table. Authors may simply choose "DROP" action to filter the malicious traffic. An instance of flow table entry is shown in Table 1.

Algorithm 6 shows the functional steps of this module. Initially, it identifies malicious devices with the help of link frequency. To achieve this, it calculates the entropy of LFIN and LFOUT to detect if there is significant amount of increase in the number of packets between any 2 end nodes in the network. Then, it compares this value with a threshold (ω) and if it is larger than it adds the corresponding device's IP into infected devices list (VIP). Finally, it identifies the controller so that the mitigation policies can be forwarded to the concerned switch for traffic sanitization.

5. Simulation & experimentation outcomes

Authors simulate SD-FCIoT network with the help of Mininet [51] simulator on a computer with Intel® Core™ i5-6600k, 3.9 GHz CPU with 16 GB of RAM, Ubuntu as operating system and a 1 TB HDD (PC2). The system is linked to the Azure Cloud service using a 32 GB RAM and 2 TB HDD (PC1). For experimental purposes, we have chosen a system with an Intel® Core™ i3-6700k, 3.2 GHz CPU, 4 GB of RAM, and a 500 GB HDD as the victim. In the following sections, authors discuss detailed network topology followed by the comprehensive evaluation study of the proposed framework.

5.1. Network topology setup

Fig. 7 shows the network topology simulated to assess the performance of the proposed framework. Here, authors implement the proposed framework on Ryu controller [52] with OpenFlow 1.5.1. There are 4 switches and each one is connected to the 4 devices.

Algorithm 6 Rules and Policies Generator

Require: Link (L), LF_{IN}, LF_{OUT}, Switch set (S), Controller set (C)
Ensure: Flow access control policies to be deployed at S

$VIP \leftarrow []$ ▷ List of victim IPs
$SM \leftarrow []$ ▷ List of malicious switches
$CM \leftarrow []$ ▷ List of malicious controllers
▷ Checking for suspicious devices
for each $l_i \in L$ **do**
  if $LF_{IN} > \omega$ **then**
    $VIP \leftarrow VIP \cup DIP$
  else if $LF_{OUT} > \omega$ **then**
    $VIP \leftarrow VIP \cup SIP$
  end if
end for
▷ Identification of malicious switches and controllers
for each $C_i \in C$ **do**
  for each $S_i \in S$ where $S_i \in C_i$ **do**
    if $V_i \in VIP \wedge V_i \in S_i$ **then**
      $SM \leftarrow SM \cup S_i$
      $CM \leftarrow CM \cup C_i$
    end if
  end for
end for
▷ Applying flow control policies at each malicious switch
for each $C_i \in CM$ **do**
  Generate the flow policy $F(SIP, DIP, SP, DP, Protocol, Action)$
  Activate F at $SM \in CM$
end for

Table 2
Distribution of instances in each attack category.

Traffic category	Instances
Benign (Normal)	34 233
Ping flood	26 530
UDP flood	16 180
HTTP flood	30 150
TCP-Syn flood	25 650
NTP amplification	24 089

Among the 20 devices, authors choose D5, D6, D10, and D11 as the malicious devices and use hping3 command for flooding attack. All 4 switches are monitored by the controller in the fog node (PC2) which is connected to the cloud server through (PC1). Devices and switches simply as acts as data forwarding nodes whereas controller is re-configured to embed the proposed framework to identify and mitigate DDoS attack. This entire setup is designed to generate the DDoS attack dataset.

5.2. Dataset details

This work aims at detecting DDoS attack traffic that originates from IoT devices inside the local network. Consumer devices have the capability to produce both normal and attack traffic simultaneously. They can also execute various forms of flooding attacks for varying durations. The dataset contains 5 distinct classes of flooding attacks including normal traffic. Table 2 represents the instances count in each attack class. Tools used for dataset comprises of hping [53] for flooding attacks, fping [54] and iperf [55] for normal traffic, and sFlow-RT for NTP amplification attack [56]. Authors use the tcpdump [57] tool to collect 'libpcap' files, which contain detailed information about network traffic packets, at the fog node. The simulation is conducted for 10 min for each attack class. Lastly, authors merge the normal traffic dataset and attack traffic dataset into a collective dataset with a total of 20 attributes, as shown in Table 3.

Authors perform data preprocessing technique such as data discretization and normalization to handle a broad range of continuous values in the dataset. Discretization is used to restrict continuous values to a limited set of values. This is accomplished to improve the accuracy of the classifiers and is carried out using the approach described in [58]. Data normalization is a process used to restrict the range of data values to a specific interval, often between 0 and 1. This is achieved by using the following equation:

$$N(a) = \frac{a - a_{min}}{a_{max} - a_{min}} \tag{10}$$

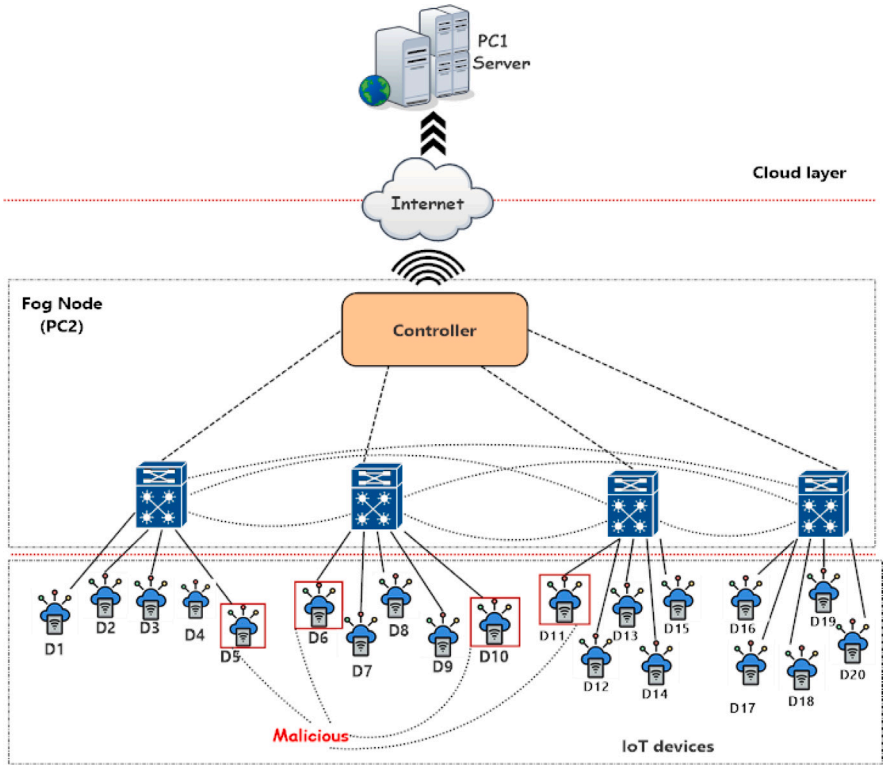


Fig. 7. Simulated Network topology.

Table 3
Original feature set in the collective database.

No.	Feature	Description
1.	ID	Packet identifier
2.	DIP	Destination IP address
3.	SIP	Source IP address
4.	S_port	Source port number
5.	D_port	Destination number
6.	#B_DIP	Number of bytes to each destination IP address
7.	Trans_rate_SIP	Packets transmitted per second from source-to-destination
8.	Trans-rate_DIP	Packets transmitted per second from destination-to-source
9.	Packet-Size	Size of packets transmitted
10.	Int_pkt_period_min	Minimum time duration between packets communicated
11.	Int_pkt_period_max	Maximum time duration between packets communicated
12.	#new_DIP	Number of new destinations approached during a session
13.	#DIP_count	Number of fixed destinations for each device
14.	SIP_PAvgR	Average transmission rate per SIP for each protocol
15.	DIP_PAvgR	Average transmission per DIP for each protocol
16.	Protocol	Network protocol used for communication
17.	#Pkt_protocol	Number of packets transmitted for each protocol
18.	#L_succ	Number of successful login
19.	#L_fail	Number of failed login attempts
20.	Packet_IN	Number of packet_IN messages

Fig. 8 depicts the significant impact of the flooding attacks on the bandwidth consumption at the controller. It may be noted that under normal situation, maximum bandwidth usage is below 50 Mbps whilst during attack scenario, this value gets double approximately.

5.3. Efficiency evaluation measures

The classifier's performance depends on the accurate design of hyperparameters, namely the selection of optimal input parameters. To accomplish this goal, the RandomizedSearchCV [59] function from the scikit-learn module in the Python programming

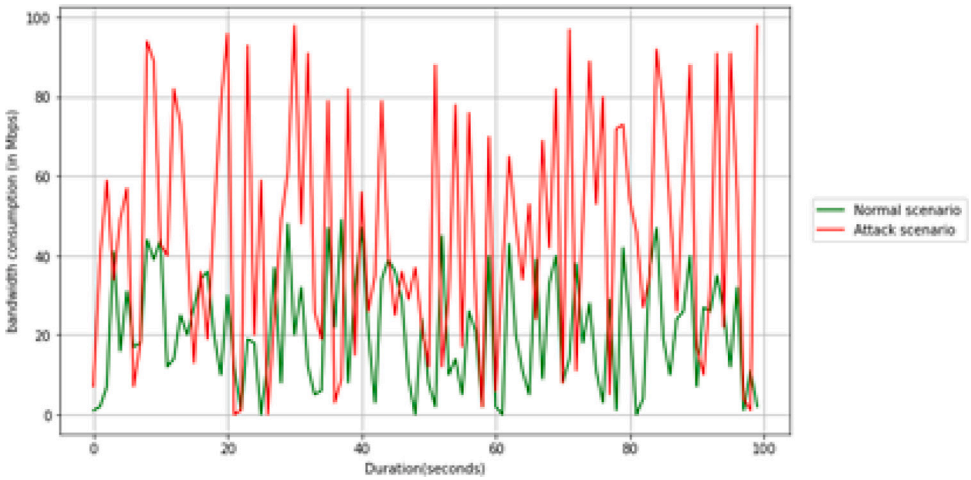


Fig. 8. Bandwidth consumption at controller in both scenarios.

Table 4
Confusion matrix.

		Predicted values	
		Attack	Normal
Actual values	Attack	True positive (TP)	False negative (FN)
	Normal	False positive (FP)	True negative (TN)

language is used. The selection is based on its ability to identify the optimal hyperparameter settings via a cross-validated search over all possible parameter values. The performance based on the following evaluation measures:

$$Recall = \frac{TP}{TP + FN} \tag{11}$$

$$FPR = \frac{FP}{FP + TN} \tag{12}$$

$$Accuracy = \frac{TP + TN}{TP + TN + FN + FP} \tag{13}$$

$$Precision = \frac{TP}{TP + FP} \tag{14}$$

$$F1 - Score = \frac{2 * Precison * recall}{Precision + Recall} \tag{15}$$

$$Misclassification\ Rate\ (MR) = \frac{FP + FN}{TP + TN + FP + FN} \tag{16}$$

These standards are determined by using confusion matrix, shown in Table 4. To perform the unbiased testing of different classifiers, all experiments are implemented through repeated k-fold cross validation (10-fold) technique. This method is selected as it reduces the variance and balances the error estimation of the validation technique [60]. In this experiment, 10 iterations of the 10-fold cross-validation procedure are completed to mitigate the bias caused by randomly selecting samples from the dataset. Final result is the average of 10 values obtained during analysis.

5.4. Results and efficiency assessment

The evaluation of the proposed framework is conducted in 2 steps. The key goal of the first step is to determine time frame (ΔT) and threshold value under varying network scenarios. The second step aims at the efficiency analysis of the 5 machine learning classifiers on the benchmark metrics, discussed in Section 5.3. Additionally, the DDoS attack detection efficiency of the proposed framework is tested on BoT-IoT dataset.

5.4.1. Parameters settings

The first step of the analysis is to precisely determine the suitable time frame to monitor the network flow for examining the current security status of the network. If the time frame is too short then we could not detect any suspicious activity because of samples shortage. Whilst, if the time frame is too long then it would take more time to respond to attack, causing delay in mitigation which leads to severe implications. Fig. 9 highlights the different time frame values and the corresponding attack detection time. It

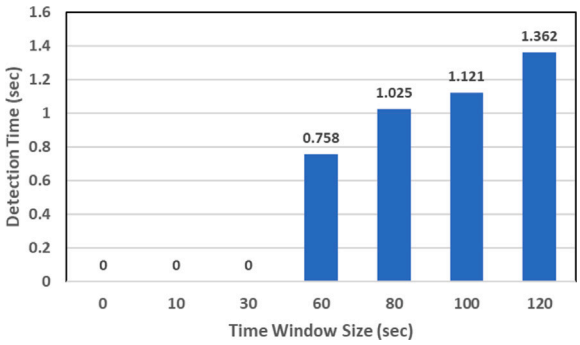
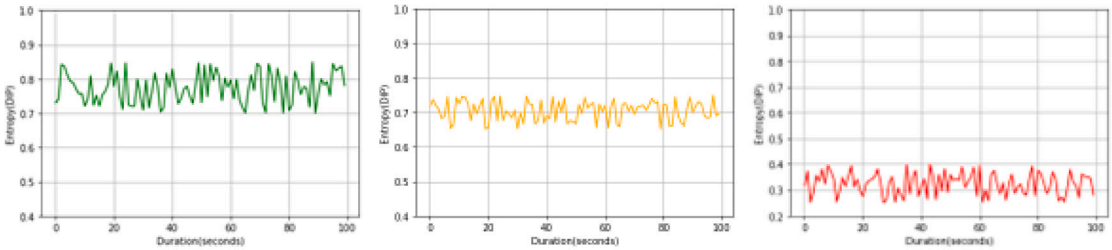


Fig. 9. Anomaly detection time for different time windows.

Table 5
Threshold value setting based on varying attack strength.

Criterion	20% malicious (w=2)	50% malicious (w=2)	80% malicious (w=1)
Mean (μ)	0.7741	0.6234	0.3751
Standard deviation(σ)	0.0132	0.0121	0.0105
$CI_{Lower}(\beta)$	0.75	0.60	0.37
$CI_{upper}()$	0.80	0.64	0.38



(a) normal traffic (b) 20% attack traffic (c) 80% attack traffic

Fig. 10. Entropy variation of DIP under different cases.

is speculated that the optimal value for time frame is 60 s as it gives shortest detection time among all the time frame in which we could be capable of detecting suspicious activities in the network.

Furthermore, if the entropy of DIP of the incoming traffic decreases below a certain threshold (β), and if this condition persists for five consecutive periods, an attack report will be generated. The Table 5 illustrates the estimated threshold value at different attack rates, based on Algorithm 3. The β value is calculated by executing Algorithm 5 times for each scenario. Fig. 10(a–c) illustrates the change in entropy of DIP under normal condition, 20% attack traffic, and 80% attack traffic scenario.

Optimized parameter values of classifiers are of vital relevance for achieving improved accuracy. This also aids in training the error penalty for stability. Table 6 provides the various parameter configurations and their respective values set using RandomizedSearchCV.

5.4.2. Efficiency analysis of classifiers

The performance assessment of the classifiers is conducted in 2 modes for both binary and multi-class classification. First mode tests the detection efficiency of each classifier on the original dataset. In the second mode, authors examine the impact of a implementing the proposed attribute selection algorithm on the detection efficacy of each classifier.

- **A. Without Attribute Selection:** Fig. 11(a)–(d) depicts the observed outcomes of different classifiers for binary classification. All parameters are set as discussed earlier in the previous sections. Note that all the findings represent the mean value of outputs obtained from 10 epochs of the 10-fold cross validation procedure. It is speculated that XGBT is the best performer with the highest detection accuracy of 99.21%, and FPR of 0.004. Furthermore, authors estimate precision, recall and F1-score to deal with dataset disparity. It is worth noticing that XGBT surpasses other classifiers as it obtains a precision of 99.61%, recall of 99.41% and f1-score of 99.51%. J48 is the least performer with the minimum accuracy of 96.01%. It may be because it generates more empty branches due to data disparity or may overfit the data.

Table 6
Configurational settings of different classifiers.

Classifier	Parameter	Explanation	Value
SVM	C	Regularization parameter	10
	γ	Controls the shape of decision boundary	0.015
	Kernel	Used for similarity checking between data points	RBF
	n_estimators	Number of trees in classifier	100
RF	min-samples_split	Minimal number of samples each internal node requires for splitting	2
	min_samples_leaf	Minimum number of samples a leaf node must retain	1
	max_depth	Maximum height of each tree in the forest	3
	max_features	Number of features to consider when looking for the best split	sqrt
	Bootstrap	Whether bootstrap samples are used when building trees	True
	Criterion	Specifies the splitting criterion in the tree	Entropy
J48	max_depth	Maximum height of the tree	3
	min_samples_split	Minimum samples required to split a node	2
	min_impurity_decrease	Minimum impurity decrease required for a split	0.0
	Prune	Whether to enable tree pruning	True
	n_neighbors	Number of neighbors used	5
KNN	Weights	Specifies each neighbor importance	Uniform
	Algorithm	Used to compute nearest neighbor	auto
	leaf_size	Leaf size for tree-based neighbor searches	30
	p	Power parameter for Minkowski distance (1 = Manhattan, 2 = Euclidean)	2
	Booster	Type of model used	gbtree
XGBT	max_depth	Maximum height of tree	3
	learning_rate	Used for reducing errors	0.1
	min_child_weight	Minimum sum of all child's observations weight	3
	Subsample	Fraction of samples used for training each tree	0.8
	colsample_bytree	Fraction of features used for training each tree	0.8
	n_estimators	Number of trees	100
	Gamma	Minimum loss reduction required to make a further partition	0.1

Fig. 12 highlights the cumulative detection latency of each classifier. It integrates the time consumed for training and testing of the classifier. It is reflected from the Fig. 12 that the maximum time is taken by XGBT and minimum time is taken by SVM. Although, XGBT is the more proficient in terms of evaluation metrices, as shown in Fig. 11, nevertheless, to choose the best competent classifier, detection latency is an indispensable measure because timely detection is crucial. Therefore, RF, which achieves second best accuracy of 98.71%, is selected as the best performing classifier with the detection latency of 120 s. Also, it can be noted that there is a small margin difference of 0.5% in accuracies of XGBT and RF.

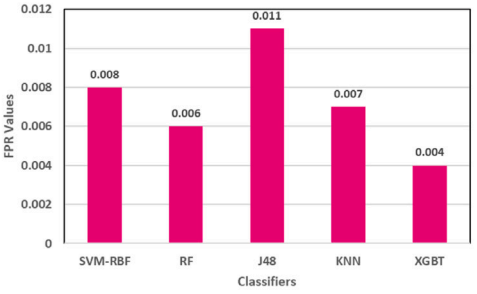
Table 7 illustrates the observed values of precision, recall, F1-score, and total time of each classifier for multi-class classification of DDoS attack. It is indicated, from Table 7, that XGBT classifier is more efficient in discriminating ping, UDP, and HTTP flood, whilst, KNN outperforms all classifiers for TCP-SYN and RF for NTP-amplification. Additionally, J48 is least scorer in all attack categories. Nonetheless, RF may be selected as the efficient classifier because it takes less time with comparable results.

B. With Attribute selection: Second mode of performance evaluation, authors assess the effect of proposed attribute selection algorithm based on symmetrical uncertainty and k-means clustering method. This algorithm returns optimal subset of attributes that helps in enhancing classifiers' accuracy. It selects those attributes that are most relevant to the class variable and no other attribute can replace them i.e., no redundant attribute (or noisy attribute). The original dataset contains a total of 20 attributes, out of which the attribute selection algorithm opts 12 most dominating attributes. The optimized dataset is then used for determining the efficacy of each classifier. Table 8 illustrates the observed experimental values of each evaluation metric.

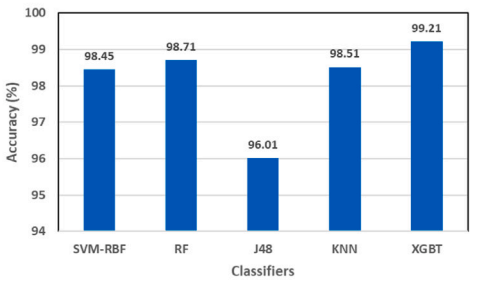
XGBT is the most efficient classifier for binary classification on the optimized dataset, achieving 99.23% accuracy (0.02% increase), FPR of 0.003, and an F1-score of 99.57% (0.06% increase). J48, though the least efficient, shows a notable 1.22% accuracy improvement. Fig. 13 illustrates ROC curves for all classifiers, where the AUC metric, ranging from 0 to 1, evaluates performance by analyzing true positive and false positive rates.

An AUC value of 0.0 indicates all predictions are incorrect, 1.0 reflects perfect predictions, and 0.5 shows no class differentiation. It is a preferred metric for standardized comparisons, independent of categorization criteria. Table 9 presents the precision, recall, and F1-score for multiclass classification on the optimized dataset, where XGBT achieves 100% accuracy for Ping Flood and TCP-SYN Flood, while RF excels in NTP Amplification with an F1-score of 97.29%.

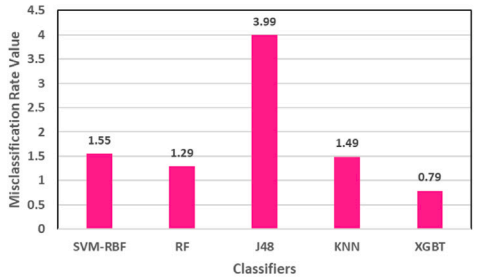
One notable observation is that KNN outperforms RF on the optimized dataset for binary classification, unlike on the original dataset. However, all classifiers show comparable performance in multiclass classification, highlighting the significant impact of the attribute selection algorithm on classifier performance.



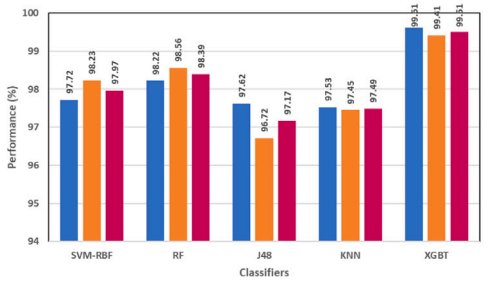
(a) FPR for binary classification



(b) Accuracy for binary classification



(c) MR value for binary classification



(d) Evaluation metrices values for binary classification

Fig. 11. Observed values of different evaluation measures of classifiers.



Fig. 12. Detection latency for binary classification.

For the timely and accurate attack detection, RF is selected as the main model in proposed attack detection phase. Additionally, more experiments were conducted to showcase the effectiveness of implementing our proposed framework in securing SD-FCIoT environment. Fig. 14 highlights the average throughput which quantifies the rate at which valid data packets are successfully sent to the IoT server in a second. It is observed during the attack in two scenarios: (1) when proposed framework is active in local controller, and (2) when no security measure is taken. According to Fig. 14, the value of throughput was lower in the absence of any security mechanism (blue line) because of the high bandwidth consumption by flooding packets, resulting in reduced capacity for the valid data packets to reach the server. Fig. 15 shows the effect of feature selection algorithm on RF's accuracy. It is worth noticing that the selection of 12–14 features contributes enhancement in detection accuracy. These features are then used for training ML classifier at the controller. An experiment was conducted to assess the benefits of fog computing in SD-IoT networks by measuring response time, which indicates the time to identify and mitigate an attack. Fig. 16 shows that the decentralized fog-based approach has a lower response time compared to the centralized approach, where a single controller at the cloud server handles traffic and distributes mitigation rules.

5.4.3. Evaluation on BoT-IoT dataset

This section discusses the attack detection efficiency of the proposed framework on BoT-IoT dataset [61] which was generated at the Cyber Range Lab of UNSW Canberra. The dataset includes DDoS, DoS, OS and Service Scan, Keylogging, and Data exfiltration

Table 7
Performance for multiclass classification.

Classifier	Attack categories	Performance measures					
		Precision	Recall	F1-score	Training time (s)	Detection time (s)	Total time (s)
SVM-RBF	Normal	100	100	100	100	90	190
	Ping-flood	97.03	98	97.51			
	UDP-flood	97.89	97.01	97.45			
	HTTP-flood	97.02	97.12	97.07			
	TCP-Syn	98.15	99.2	98.67			
	NTP amplification	96.2	97.45	96.82			
RF	Normal	100	100	100	95	80	175
	Ping-flood	98.23	98.32	98.27			
	UDP-flood	97	97.25	97.12			
	HTTP-flood	98.01	98	98			
	TCP-Syn	98.89	98.36	98.62			
	NTP amplification	97.51	98	97.75			
J48	Normal	99	100	99.5	120	100	220
	Ping-flood	97.32	96.5	96.91			
	UDP-flood	97.12	97.01	97.06			
	HTTP-flood	96.99	96.59	96.79			
	TCP-Syn	97.56	97.23	97.39			
	NTP amplification	96	96.02	96.01			
KNN	Normal	99	100	99.5	180	150	330
	Ping-flood	98	97.35	97.67			
	UDP-flood	97.89	98.5	98.19			
	HTTP-flood	97.79	97.12	97.45			
	TCP-Syn	100	100	100			
	NTP amplification	94	95	94.5			
XGBT	Normal	100	100	100	250	140	390
	Ping-flood	98.65	98.88	98.76			
	UDP-flood	98.23	97.96	98.09			
	HTTP-flood	98.89	98.9	98.89			
	TCP-Syn	100	99.65	99.82			
	NTP amplification	97.56	97.65	97.6			

Table 8
Performance for binary classification.

Classifiers	Accuracy	FPR	MR	Precision	Recall	F1-score
SVM-RBF	98.23	0.007	1.77	98.19	97.1	97.64
RF	98.89	0.006	1.11	98.23	97.25	97.74
J48	97.23	0.009	2.77	98	97	97.5
KNN	99	0.005	1	98.45	98.89	98.67
XGBT	99.23	0.003	0.77	99.52	99.62	99.57

threats, with DDoS further classified into 3 classes comprising TCP, UDP, and HTTP flood. RF is implemented with our attribute selection algorithm to distinguish the DDoS attack classes. Table 10 highlights the observed results for binary classification. Here, an accuracy of 99.61% is achieved with FPR value of 0.003 and precision of 98.89%. Table 11 reveals the results for multiclass classification of DDoS attack. It is clearly reflected that RF is attaining higher accuracy of 99.06% and 99% for HTTP and TCP, respectively

5.4.4. Comparative analysis

Several research efforts have been made in the field of DDoS detection and mitigation, each employing different methodologies to address the challenges posed by such attacks. Considering these contributions, a comparative analysis has been performed to evaluate the efficiency and effectiveness of the proposed approach. This analysis helps to clearly highlight where our approach stands in relation to existing solutions in terms of key factors such as detection accuracy, scalability, and resource usage. Table 12 summarizes the salient features of previous works and contrasts them with the proposed approach, emphasizing their respective advantages and limitations.

6. Conclusion

This study presents a robust multi-phase framework for detecting and mitigating DDoS attacks in SD-FCIoT networks. The proposed framework leverages entropy-based anomaly detection for early identification and machine learning models for precise attack classification, supported by a novel attribute selection algorithm utilizing symmetrical uncertainty and k-means clustering. The dynamic thresholding mechanism ensures adaptability to varying network conditions, enhancing detection accuracy while

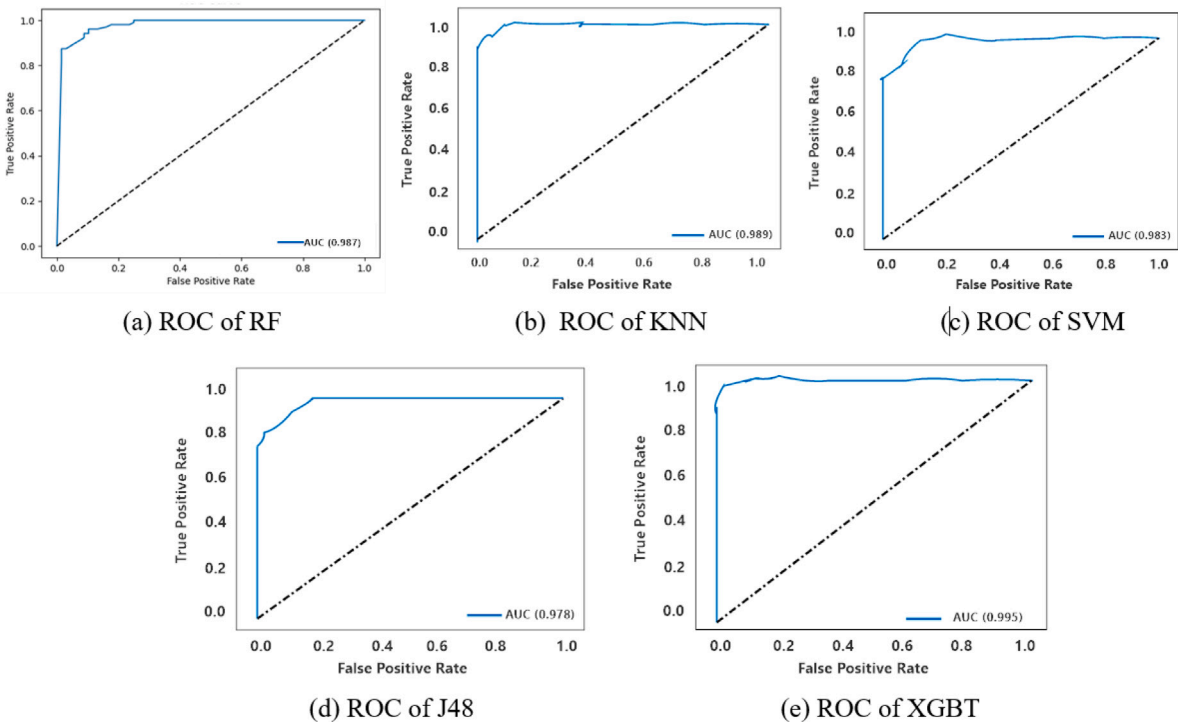


Fig. 13. ROC curves of different classifiers.

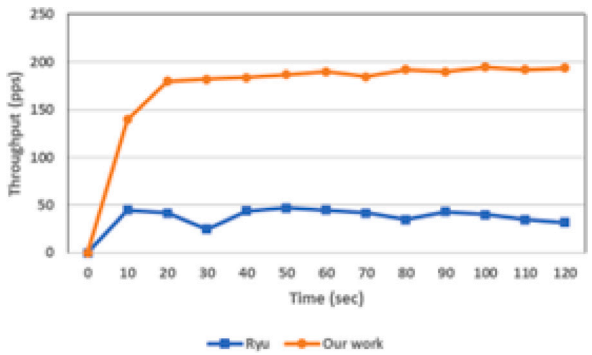


Fig. 14. Throughput variation.

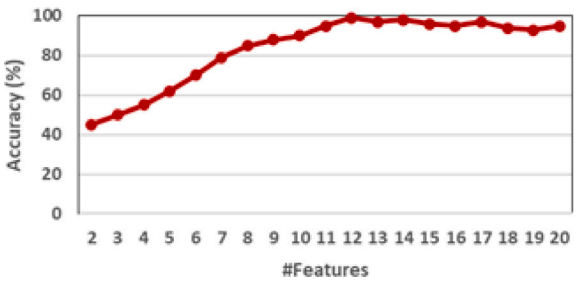


Fig. 15. Accuracy variation using attribute selection.

minimizing false positives. Comprehensive experiments on a simulated SD-FCIoT environment and the BoT-IoT dataset validate the framework's efficiency. Random Forest emerged as the most effective classifier, achieving 98.71% accuracy in binary classification

Table 9
Performance for multiclass classification.

Classifier	Attack categories	Performance measures		
		Precision	Recall	F1-score
SVM-RBF	Normal	100	100	100
	Ping-flood	98	97.85	97.92
	UDP-flood	98	97.89	97.94
	HTTP-flood	97	98	97.5
	TCP-Syn	99	99	99
	NTP amplification	97	97	97
RF	Normal	100	100	100
	Ping-flood	98.32	98.3	98.31
	UDP-flood	97.5	97.12	97.31
	HTTP-flood	98	98.11	98.05
	TCP-Syn	99	99	99
	NTP amplification	96.89	97.7	97.29
J48	Normal	100	100	100
	Ping-flood	97.35	97	97.17
	UDP-flood	97.22	97.32	97.27
	HTTP-flood	97	97.45	97.22
	TCP-Syn	98	98.2	98.1
	NTP amplification	96.52	96.82	96.67
KNN	Normal	100	100	100
	Ping-flood	98.25	98.42	98.33
	UDP-flood	98	98.2	98.1
	HTTP-flood	98	97.65	97.82
	TCP-Syn	99	99	99
	NTP amplification	96	96.13	96.06
XGBT	Normal	100	100	100
	Ping-flood	100	100	100
	UDP-flood	99	97.5	98.24
	HTTP-flood	98.85	99	98.92
	TCP-Syn	100	100	100
	NTP amplification	98.5	98.52	98.51

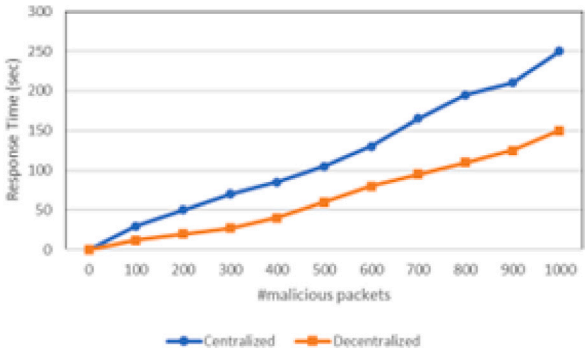


Fig. 16. Response time variation.

Table 10
Performance for binary classification using RF+ AS.

Metrics	Performance (%)
Accuracy	99.61
FPR	0.003
MR	0.39
Precision	98.89
Recall	99
F1-score	98.94

Table 11
Performance matrix using RF+ AS for DDoS sub-categorization.

Attack categories	Performance (%)				
	Accuracy	FPR	Precision	Recall	F1-score
Normal	100	0	100	100	100
DDoS-TCP	98.78	0.005	99	99	99
DDoS-UDP	99.32	0.004	98.9	98.93	98.91
DDoS-HTTP	98.98	0.006	99	99.12	99.06

Table 12
Comparative analysis of the proposed approach with existing work.

Study	Detection approach	Advantages	Limitations	Proposed approach
Ezeme et al. [62]	Entropy + PCA	High detection rate with improved recall and F1-score using Tsallis entropy.	Static threshold values; limited adaptation to dynamic network traffic.	Dynamic entropy-based anomaly detection with threshold adjustment using Chebyshev's inequality.
Santaniello et al. [63]	Entropy-based fog layer IDS	Early detection at the fog layer reduces detection latency.	Relies on static entropy thresholds; vulnerable to fluctuating traffic conditions.	Combines dynamic entropy and machine learning for robust, real-time detection and classification.
Ali et al. [64]	Entropy + Sequential probability	Improved precision by combining entropy with probabilistic models.	Fixed thresholds limit detection flexibility during network fluctuations.	Dynamic thresholds with multi-phase mitigation ensure flexibility and accuracy in diverse scenarios.
Kumar et al. [65]	Federated learning + Random forest, XGBoost	Distributed detection reduces training latency and enhances scalability.	High computational overhead and resource consumption for federated learning in resource-constrained devices.	Lightweight framework integrating statistical methods and ML models optimized for edge-level IoT devices.
Mishra et al. [66]	Entropy in SDN-Cloud environment	High detection rate (98.2%) in centralized SDN-based IoT environments.	Centralized approach suffers from latency and scalability issues in large-scale IoT deployments.	Decentralized fog-enabled SDN architecture for low-latency, scalable, and near-end-node detection.
Roopak et al. [67]	Deep learning (LSTM, CNN)	Effective detection of complex attack patterns with high accuracy.	High computational complexity unsuitable for real-time edge-based IoT environments.	Lightweight machine learning models (e.g., RF, XGBoost) for efficient and timely classification.
Proposed framework	Dynamic entropy + Machine learning	Combines statistical and ML-based techniques for robust, real-time detection.	Requires training and fine-tuning ML models; dependent on fog node processing capabilities.	Multi-phase framework with dynamic thresholds, feature selection, and real-time mitigation policies.

and high performance in multi-class detection. The attribute selection algorithm improved accuracy by up to 1.22%, and the fog-enabled architecture significantly reduced response time by 35% compared to centralized systems. The framework's ability to provide scalable, low-latency, and accurate DDoS defense highlights its practical applicability in securing CIoT networks. Future work will focus on extending the framework to identify a broader range of attack types and incorporating advanced detection techniques for enhanced adaptability in evolving IoT environments.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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